NUNA 370 Townsend St. San Francisco, CA 94107

Attention: CMS-0042-NC

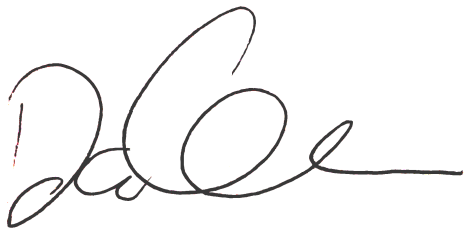
Dr. Mehmet Oz
Administrator
Centers for Medicare & Medicaid Services
Department of Health and Human Services
P.O. Box 8013,
Baltimore, MD 21244-8013

Dear Dr. Oz,

I write on behalf of Nuna, Inc., a digital health company that shares your vision of a healthcare system that empowers Americans to live their healthiest & happiest lives **and** delivers the care they want, need, and deserve.

At Nuna, we're inspired by CMS's ambition to foster a vibrant and interconnected "Health Technology Ecosystem" and believe it marks a pivotal moment in advancing the nation's healthcare technology infrastructure. The strategies contemplated in RFI CMS-0042-NC have the potential to help Nuna, and hundreds of other digital health companies like us, deliver better care for beneficiaries across the United States.

We thank CMS and the ONC for the opportunity to comment on the RFI and are pleased to offer these recommendations based on our experience in the field.



David Chen
Co-Founder and Chief Data Officer, Nuna, Inc.

Part 1: Promote Data Interchange Standards To Enable Health Impact @ Scale

At Nuna, data is the foundation of our products. However, we've found that it's nearly always stored in different formats from site to site which greatly limits our ability to scale without significant integration work. By driving adoption of data standards & APIs, CMS could greatly enhance the ability of digital health solutions to efficiently scale to millions of beneficiaries (TD-9.a). In practice we suggest CMS:

Require EHRs to **first** expose data necessary to enroll patients in digital health solutions

Since there's no "one-size-fits-all" solution for exposing clinical data via APIs, we suggest CMS incentivize providers to expose high-quality data that first helps identify patients who would benefit from digital health solutions and then expand the scope of supported data over time (PR-3.e):

- **First**: Ensure EHR APIs are able to share clinical data necessary to establish a patient's need for health management. This will enable digital health solutions to identify eligible patients for enrollment. For example, ensure that EHRs are able to expose clinical records, diagnoses, or lab-measure values that identify a patient as having unmanaged type 2 diabetes (TD-2.b).
- **Next**: Incent adoption & implementation of FHIR-based APIs that enable approved digital health solutions to push a limited scope of data back to the EHR **if the provider organization wants to import that data**. This will ensure that the EHR, as a clinical system-of-record, always has the most up-to-date data for a patient.
- **Later**: Incent adoption & implementation of standardized APIs that support FHIR-based export of a broader spectrum of data required to better manage a target condition of interest (e.g. beyond USCDI v3). CMS can prioritize this data based on feedback from solution developers, on a per-condition basis.

Incent & enable **bulk** sharing of patient data to delegated entities

Nuna's solutions require bulk access to patient data to enroll patients at scale. We recommend that CMS support delegated access to bulk data for its APIs and require similar design patterns for EHR APIs. In practice CMS should:

- Enable a provider to create, manage, and share access credentials that enable one or more digital health vendors to pull data for a subset of aligned beneficiaries using the BCDA. This would enable vendors to efficiently identify all patients with qualifying conditions. (PC-8.a, PC-9.a, TD-2.a, TD-15.a)
- Generalize the BCDA to enable bulk export of claims data for all beneficiaries seen by a provider organization, independent of the covered payment model.
- Work with HL7 to mature the FHIR bulk-data implementation guide to `normative` status**,** from `Trial-use`, and specify FHIR as the export format required for ONC Health IT Certification.
- Require EHR vendors to implement OAuth 2.0 access management for bulk data exports which would make it easier for providers to delegate bulk-data access to digital health vendors.
- Specify reasonable performance requirements for bulk-data APIs (PC-5.b) as many implementations to date have been too slow to be operationally useful (e.g. took days to generate exports).

Incent **digital health solutions** to share patient data using standard APIs

As Nuna has brought its RPM solutions to market, we've commonly found that provider organizations are already trying to juggle multiple RPM solutions (e.g. one for COPD, one for diabetes, etc.), and the cognitive burden of adopting yet another proprietary interface is simply too high. We see this problem continuing to grow in the near future: As the market grows to offer hundreds of diverse and competitive solutions, it will be impossible for physicians to adapt to and adopt dozens of proprietary product interfaces (PR-1, PR-1.a). To address this issue, we suggest digital health solutions make relevant patient data available to providers via standardized APIs. This will enable providers to import data of interest from numerous solutions into a single location, paired with a unified user experience (look-and-feel and feature set), one-off EMR integration, and a single implementation of the clinical team care access model. In practice, CMS should:

- Require digital health solutions to share provider-facing patient data via standardized APIs as a condition for app certification.
- Require digital health solutions to establish and maintain endpoints that enable data access for provider products that may consume and display data from many digital health solutions.
- Study and promote provider experience platforms and their connection to EMRs and clinical workflow

In the end, digital health solution vendors should see these recommendations as helpful. The vendors can focus on patient experience and their product's value proposition, while availing themselves of a network effect that solves backend obligations.

Part 2: On Encouraging The Adoption of Digital Health Solutions

To date, Nuna has identified many points of friction that can discourage users from adopting and using our products. By addressing some of these issues up front, CMS could increase the adoption rate of digital health solutions at large. In practice we suggest CMS:

Make it easy for beneficiaries to enroll in and import health data into apps

Industry data shows that use of SSO decreases the friction to app adoption (TD-3.a). Furthermore, use of a single patient digital identity would improve patient safety. CMS should encourage beneficiaries, app developers, providers, and payers to make use of a single digital identity across digital health apps (PC-6, PC-14.b). In practice CMS should:

- Identify a preferred single-sign-on service that can be used for any certified app. This service should provide common information about a beneficiary, including their MBI, a registry of a patient's data sources, and authorize apps to pull patient data from CMS and EHRs. Alternatively, consider enabling medicare.gov to be used as an SSO service for apps.
- Require EHRs to share clinical data for SSO-authenticated users, keyed on the user's MBI. This will make it easier to fetch a user's data across EHRs (PR-9, PR-10.a)
- Consider developing extensions to this framework to work for MA beneficiaries as well.

Make it easy for beneficiaries to identify & adopt the right solutions for them

For generally available solutions (i.e. without physician referral), it's in the best interest of beneficiaries, CMS, and vendors to ensure that beneficiaries can adopt effective apps. In practice CMS should:

- Publish a single directory of certified apps where patients can find trusted solutions (PC-5.a).
- Work with App Store operators (i.e. iOS store, Google Play store) to "badge" certified apps.
- Require certified apps to publish standard labels (similar to an FDA nutrition label or FCC broadband Facts) to help patients compare and contrast solutions. Labels should convey common and helpful properties of each solution (e.g. who it is for, how it works, expected effect, rating) (PC-7).
- Require apps to use a patient's data to confirm the patient is a target user when applicable.
- Enable direct-to-vendor funding arrangements so that patients don't have to provide payment to gain access to solutions. Requiring direct payment from users will likely reduce rates of solution adoption.

Review and certify safe & effective solutions

Patients generally distrust tech companies when it comes to healthcare, presenting meaningful barriers to the adoption of digital health solutions. Furthermore, physicians themselves harbor skepticism of the value digital solutions bring to the table. CMS can help rebuild that trust by operating an evaluation and certification process that helps ***patients and providers*** *identify safe and effective digital health solutions (PC-5.a). In practice CMS should:*

- Develop or identify best-practice guidelines for developing safe, secure apps (e.g. OWASP MASVS).
- Develop or identify best-practices for evaluating app effectiveness, including appropriate use of real-world evidence.
- Review and certify apps that are:
  - **Safe:** the solution is safe and/or poses minimal risk to the patient, including the safe use of generative AI where applicable [details].
  - **Effective:** The vendor has evaluated the solution using best practices and results show evidence of effectiveness.
  - **Secure:** The app meets requirements for the protection and appropriate use of patient data.
  - **Sovereign:** The app serves the patient first and foremost; The app will not share any patient health data unless explicitly allowed by the patient, and the vendor will never sell user data.
- Catalog and "tag" solutions that meet the standard of being "Primarily Health-Related" or "Meets reasonable expectations of improving or maintaining the health or overall function." This could streamline adoption of digital health solutions by Medicare Advantage Organizations [details].
- Consider assessing app effectiveness using a set of patient-centric, quality-of-life metrics, similar to those used by the DiGA model.
- Work with the FDA to establish clear and more appropriate thresholds for determining when a solution should qualify as a regulated medical device. By aligning oversight with actual risk, CMS can reduce development overhead and support responsible innovation.

Provide guidance on the safe and effective use of AI

Amid a patchwork of evolving state-level AI and digital health regulations, CMS should strive to establish a federal foundation that helps developers navigate compliance and align with national standards for safe and effective AI use, while lowering developer costs associated with regulatory overhead and preserving flexibility that encourages innovation.

- CMS should identify specific high-needs, low-risk innovation pathways and 1) give developers a safe-harbor framework under which to create new solutions with minimal regulatory overhead and 2) promote use of real-world evidence to validate those solutions.
- Consider developing a companion healthcare-specific risk profile, aligned with the NIST AI Risk Management Framework, that identifies key risks (e.g., patient harm, data misuse, model inaccuracy or drift) and corresponding mitigation and monitoring strategies.
- Enhance or supplement CMS's Acceptable Risk Safeguards (ARS) 5.1 to include AI-tailored safeguards to address risks such as data drift, reduced model accuracy, and model explainability.

Part 3: Modernize Payment Policy For The Digital Health World

The opportunities to enable better health using digital solutions are significant, but the risks and costs of developing said solutions are also high, ultimately tempering industry interest and investment. To fully catalyze industry engagement, CMS should formulate clear & durable payment streams tailored for the longitudinal impact potential of digital health solutions (TD-1). In practice, CMS should:

Augment **existing** payment streams to foster adoption of digital health solutions **today**

- Allow services reimbursed under a remote monitoring model to credit staff time under "General Supervision" instead of "Direct Supervision" where feasible. This will give provider organizations more operational flexibility when supporting digital health solutions.
- Provide additional detail and guidance on the requirements that make a digital health solution "fundable" using MA rebates and/or MSSP payment streams (shared savings, AIPs) (VB-1).
- Consider reducing the amount of justification that an MA plan must provide in support of offering a certified digital health solution as a supplemental benefit. For example, remove the requirement that a plan submit studies of solution effectiveness as effectiveness will have been established during the certification process.
- Consider treating expenditures on digital health solutions as "quality improvement expenditures" for the purposes of MLR calculation (e.g. count towards MLR numerator rather than denominator).
- Adjust CMMI's value-based payment models such that hospital systems are rewarded for improvements in care, even if the patient is attributed to an independent PCP outside the system. For example, if a hospital spurs a patient to adopt a digital health solution, at the time of discharge, that improves outcomes (e.g. fewer readmissions and lower total cost of care), the hospital should share in the upside performance bonuses, even if the patient is attributed to an independent physician.

Develop payment streams that support direct-to-beneficiary relationships, **independent of insurance**, to foster focus on long-term health

On average, patients change MA plans every ~3.5 years. Consequently, payers and providers tend to adopt solutions that deliver patient benefits (and business value) in <3 years while ignoring solutions that deliver better outcomes on a longer timeline. CMS ultimately bears the financial burden of this short-term focus.

*To enable better long-term patient health, CMS should consider mediating a direct relationship between patients and digital health solutions, **independent of the patient's insurance and/or provider**. This would enable certain digital health solutions to focus on long-term patient health as the relationship would persist, even if the patient changes provider or insurance.*

Enabling a direct-to-beneficiary relationship would represent a meaningful change in the way CMS operates and interacts with beneficiaries and vendors; we identify multiple strategies (Table 1) for the sake of consideration but with full appreciation that some paths may be logistically or technically impractical.

Table 1: Strategies For Enabling Direct-To-Beneficiary Relationships

App Access Strategy: CMS gives beneficiaries "Wellness-wallet" credit that can be used to license certified apps	
Payment Policy	*Benefits And Challenges Of Approach*
CMS Pays *Prospectively* for User Engagement With Certified Apps - CMS allocates beneficiaries a fixed amount (e.g. $5 pmpm) that can be used as credit towards any certified app. - CMS could offer different amounts of credit for different health benefits (e.g. $5 pmpm for wellness, $10 pmpm for chronic condition management) - Beneficiaries adopt certified apps of their choosing. - CMS pays vendors monthly for achieving requisite levels of actions or engagement.	- (Pro) Lets beneficiaries pick and choose solutions that work for them. - (Pro) Works for a broad-spectrum of self-managed health activities, not just chronic condition management. - (Pro) Provides clear, direct incentives to attract business and developer interest. - (Con) If app certification standards are not rigorous, would be an avenue for fraud/waste/abuse.
App Access Strategy: CMS pays digital health solutions for delivering improved beneficiary outcomes	
Payment Policy	*Benefits And Challenges Of Approach*
CMS Pays *Retrospectively* for Outcomes Delivered By Apps - CMS names indications and outcomes it wants to purchase, and price points for delivering those outcomes. - Eligible patients adopt certified apps of their choosing. - Apps measure patient outcomes and report to CMS to receive payment.	- (Pro) Lets the market select for effective solutions. - (Pro) Best stewardship of CMS funds: payment only made if outcomes are realized. - (Con) Long timelines and uncertain payments for vendors. - (Con) Opens the door to user-shopping by vendors. - (Con) Challenging to operationalize.
App Access Strategy: CMS offers premium incentives for beneficiaries; Beneficiaries license solutions directly	
Payment Policy	*Benefits And Challenges Of Approach*
CMS offers incentives to beneficiaries for improving their health - CMS identifies indications and outcomes it wants to purchase, and offers incentives to beneficiaries for achieving those outcomes. For example, CMS could offer incentives to beneficiaries for completing cancer screens or annual wellness visits, for monitoring blood pressure regularly if they have hypertension, or for being med-adherent. - Available incentives are published directly to beneficiaries. - Beneficiaries directly identify and license apps they think will help achieve the target outcomes. - If a beneficiary achieves a target outcome, regardless of app usage, they receive the reward.	- (Pro) Directly incentivizes beneficiaries to complete actions under their control rather than paying a provider to try and convince a beneficiary to complete an action. - (Pro) Follows a standard B2C model, without dependency on CMS to fund apps. - (Pro) Lets the market select for effective solutions. - (Con) Requires careful development of policy that won't penalize beneficiaries who are unable to improve their health (perhaps beneficiary is already in good health or has an unaddressable condition). - (Con) Challenging to operationalize.

Conclusions

In closing, Nuna is highly optimistic about CMS's vision for a thriving digital health ecosystem, and hope the tactical aspects discussed in this response will be helpful in making this vision a reality. If at all helpful, we welcome any opportunity to further discuss or develop these crucial topics with CMS, providers, patients, and other industry representatives.